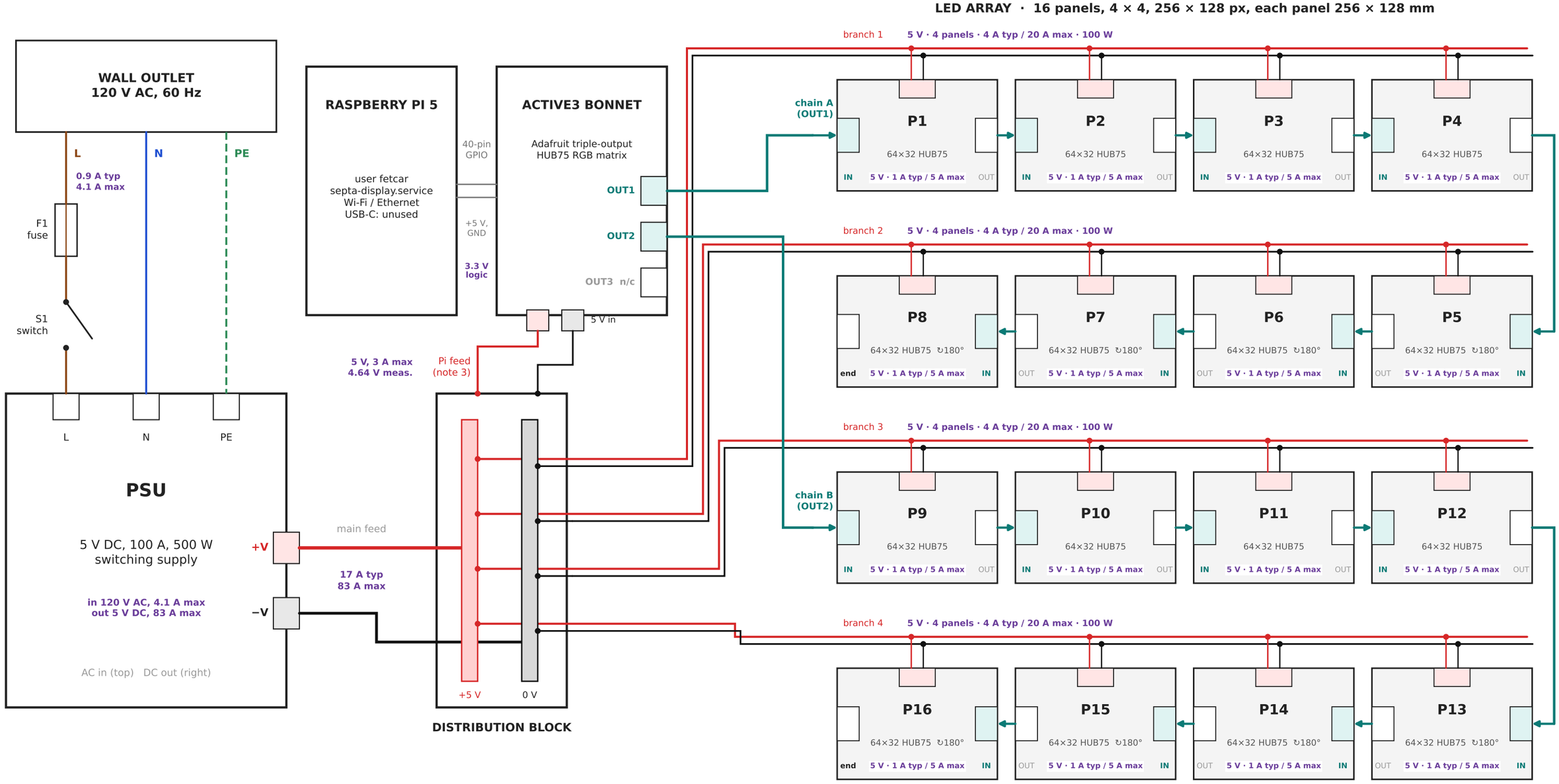


LEGEND

- +5 V DC
- 0 V / GND
- HUB75 ribbon, 5 V logic, arrow = data direction
- mains line (L)
- mains neutral (N)
- protective earth (PE)
- stacked 40-pin header
- 5 V · 20 A — rating (see table)
- junction

NOTES

1. Load: about 1 A per panel typical, 5 A at full white (E90 measurement). 16 panels → 80 A worst case, about 400 W. PSU rated 100 A. See ratings table.
2. One wall outlet. Mains passes through F1 and S1 in the PSU cord. The fuse rating was not recorded; label it when found.
3. The Pi takes 5 V from the PSU through wires soldered to the bonnet's power input, then the 40-pin header. Its USB-C input is unused. Measured 4.64 V at the Pi under full white (under-voltage limit 4.63 V). Fix: dedicated 16 AWG pair from the PSU terminals; see HARDWARE.md §1.
4. Two chains of eight on bonnet outputs 1 and 2; output 3 unused. One chain of sixteen refreshed at about 30 fps and flickered (E90 report, Methods I).
5. Serpentine: rows 2 and 4 are mounted turned 180°, so each chain doubles back. build_map() and combine_maps() in swarthmore-tracked.py encode this wiring; recabling the array needs a matching software change or the picture scrambles into blocks.
6. Panel positions are logical, as in the original sketch. Before rewiring, confirm on the hardware: which output feeds the top half (the code assumes OUT1), which corner each chain enters, whether the Pi feed leaves the block or the PSU terminals, and conductor gauges.
7. Supersedes the May 2026 hand sketch (E90 report Fig. 7), which showed a second outlet for the Pi and one 16-panel chain.

ELECTRICAL RATINGS

Circuit	Voltage	Typ	Max	Power	Wire*
Mains in, L / N	120 V AC, 60 Hz	0.9 A	4.1 A	490 W	14 AWG
PSU output (rated)	5.0 V DC		100 A	500 W	terminals
Main feed, PSU → block	5 V DC	17 A	83 A	415 W	4 AWG
Each branch, 4 panels	5 V DC	4 A	20 A	100 W	12 AWG
Each panel	5 V DC	1 A	5 A	25 W	panel lead
Pi + bonnet feed	5 V (4.64 meas.)	1 A	3 A	15 W	16 AWG
GPIO, Pi → bonnet	3.3 V logic				header
HUB75 ribbons	5 V logic				16-way IDC

typ = normal board, max = all white. Mains at 85 % PSU efficiency. Under-voltage at 4.63 V.

* Recommended minimum copper for runs under 1 m; as-built gauges not yet verified (note 6).

SEPTA SWARTHMORE STATION DISPLAY

System wiring diagram, as built

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Advisor: Prof. E. Masroor · Swarthmore College Engineering

Maintainer / drawn by: B. Greenberg

github.com/BenGreenberg07/septa-tracker

DATE	REV
2026-10-21	B
SCALE	SHEET
None (schematic)	1 of 1